

Car Rental CRM Project – Full Explanation

Phase 1: Problem Understanding & Industry Analysis

👉 **Goal:** Understand what we're building and why.

Requirement Gathering

- Talk to stakeholders (car rental company manager, rental agents, customer service).
- Example requirement:
 - Track all cars (`Car__c`) with availability status (`Car_Status__c`).
 - Allow customers (`Contact`) to book cars (`Car_Experience__c`).
 - Prevent overlapping bookings.
 - Generate revenue reports.

Stakeholder Analysis

- **Admin:** Managing system setup
- **Rental Agents:** Create/manage bookings
- **Manager:** Approves high-value rentals, monitors reports
- **Customer Service:** Handles issues/returns

Business Process Mapping

Customer requests booking → Agent checks availability → Booking created → Approval (if high value) → Email notification.

Industry-specific Use Case Analysis

- Cars have limited availability.
- Rates vary by model/season.
- Maintenance tracking is important.
- Need fleet tracking + automation + notifications.

AppExchange Exploration

- Some apps exist → but you build your custom solution for learning.
-

Phase 2: Org Setup & Configuration

👉 **Goal: Prepare Salesforce environment.**

- Use **Developer Edition Org**
- Set up **Company Profile**, time zone, currency
- Set **Business Hours & Holidays**
- Fiscal year → Standard (Jan–Dec)
- Create users (Agent, Manager)
- **Profiles** → Agents (restricted), Manager (full access)
- **Roles** → Manager → Agent
- **Permission Sets** for extra access
- **OWD**
 - Cars → Public Read Only
 - Bookings → Private
- **Sharing Rules** (if needed)

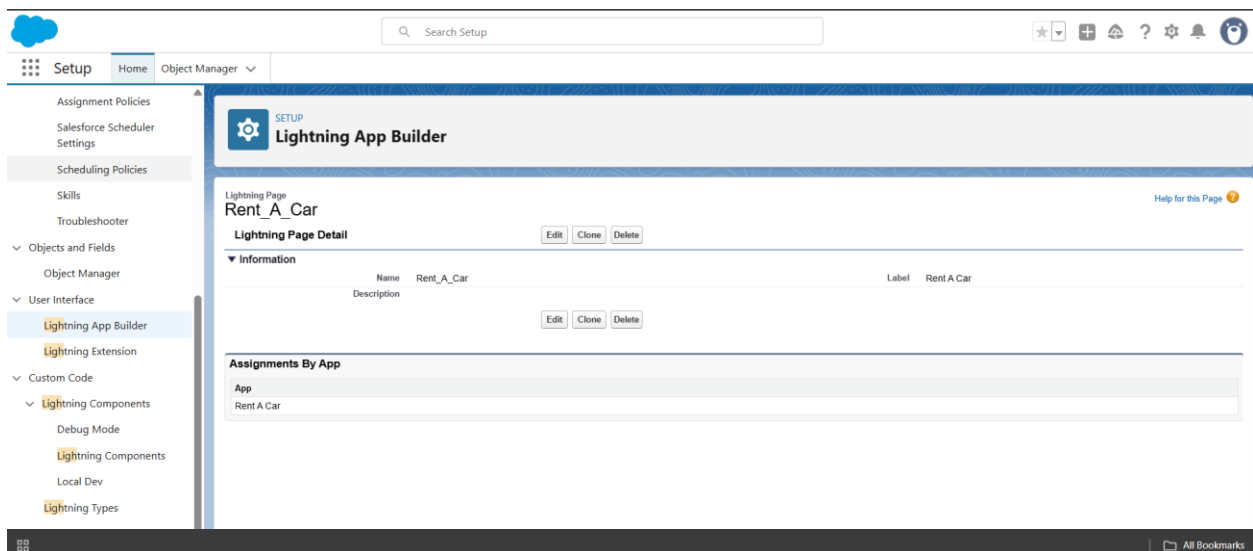
- Restrict **Login Hours**
- Build in **Sandbox**, deploy using **Change Sets**

Phase 3: Data Modeling & Relationships

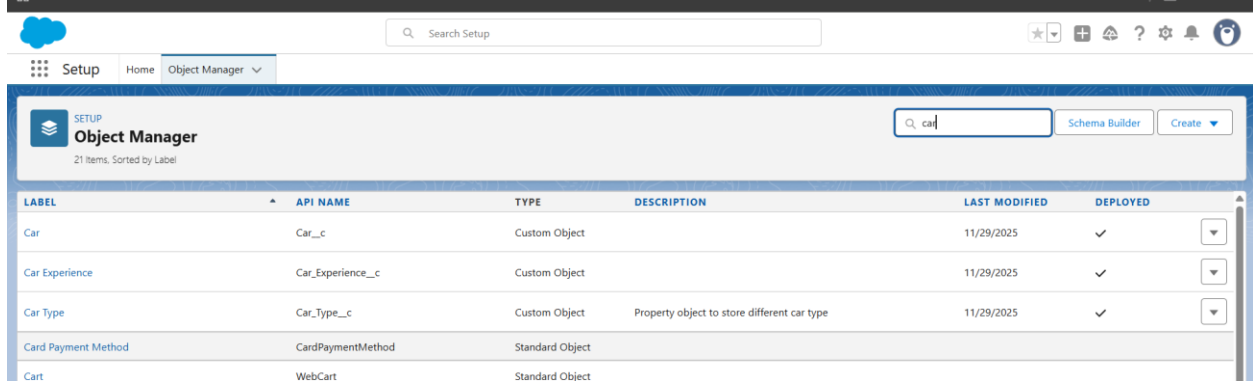
👉 **Goal: Create Salesforce data structure.**

Objects

- **Standard:** `Contact`
- **Custom:** `Car__c`, `Car_Experience__c`



The screenshot shows the Salesforce Lightning App Builder interface. The left sidebar contains a navigation menu with options like Setup, Home, Object Manager, and various configuration settings. The main content area displays the 'Lightning Page Detail' for the 'Rent A Car' app. It includes a section for 'Information' with fields for Name (Rent A Car) and Label (Rent A Car). Below this is an 'Assignments By App' section showing the app 'Rent A Car'.



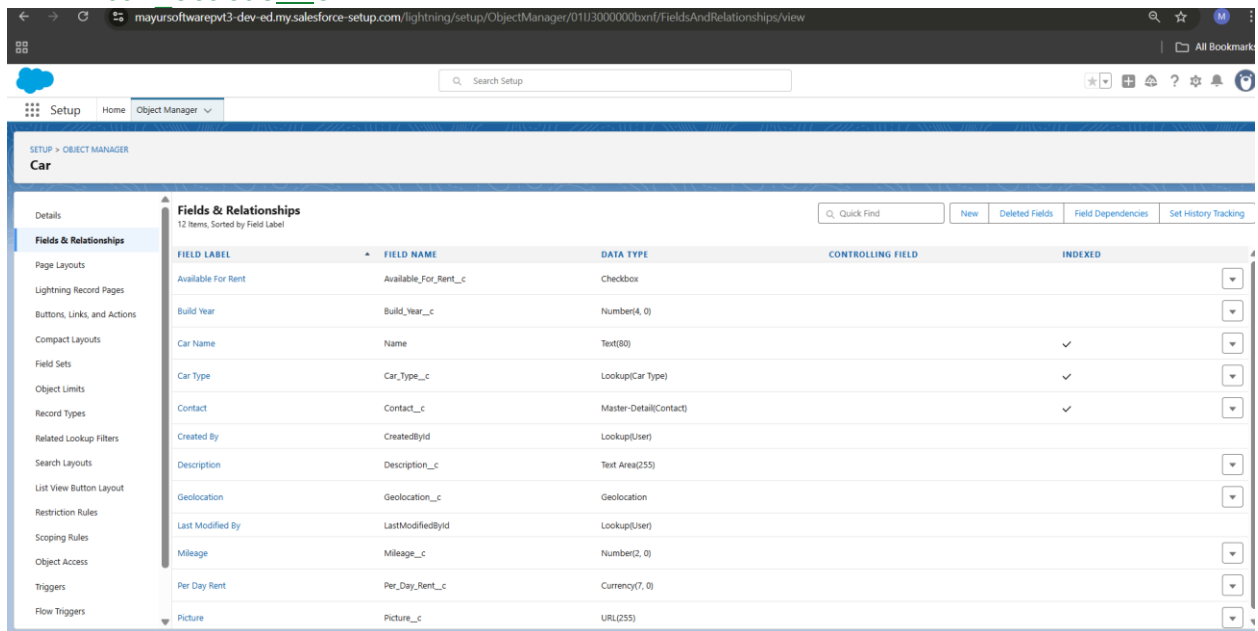
The screenshot shows the Salesforce Object Manager interface. It displays a list of objects with columns for Label, API Name, Type, Description, Last Modified, and Deployed. The objects are sorted by Label.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Car	Car__c	Custom Object		11/29/2025	✓
Car Experience	Car_Experience__c	Custom Object		11/29/2025	✓
Car Type	Car_Type__c	Custom Object	Property object to store different car type	11/29/2025	✓
Card Payment Method	CardPaymentMethod	Standard Object			
Cart	WebCart	Standard Object			

Fields

Car__c

- Car_Model__c
- Car_Registration_No__c
- Car_Daily_Rate__c
- Car_Status__c



The screenshot shows the Salesforce Setup interface for the 'Car' object. The 'Fields & Relationships' section is active, displaying a list of 12 fields. The table below represents the data shown in the screenshot.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Available For Rent	Available_For_Rent__c	Checkbox		
Build Year	Build_Year__c	Number(4, 0)		
Car Name	Name	Text(80)		✓
Car Type	Car_Type__c	Lookup(Car Type)		✓
Contact	Contact__c	Master-Detail(Contact)		✓
Created By	CreatedByid	Lookup(User)		
Description	Description__c	Text Area(255)		
Geolocation	Geolocation__c	Geolocation		
Last Modified By	LastModifiedByid	Lookup(User)		
Mileage	Mileage__c	Number(2, 0)		
Per Day Rent	Per_Day_Rent__c	Currency(7, 0)		
Picture	Picture__c	URL(255)		

Car_Experience__c (Booking)

- Booking_Start_Date__c
- Booking_End_Date__c
- Booking_Total_Amount__c

mayursoftwarept3-dev-ed.my.salesforce-setup.com/lightning/setup/ObjectManager/0113000000bxnd/FieldsAndRelationships/view

Setup

Object Manager

Search Setup

Setup > OBJECT MANAGER

Car Experience

Details

Fields & Relationships

6 Items, Sorted by Field Label

Quick Find

New

Deleted Fields

Field Dependencies

Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Car	Car__c	Lookup(Car)		✓
Created By	CreatedById	Lookup(User)		
Experience	Experience__c	Rich Text Area(32768)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Title	Name	Text(80)		✓

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

List View Button Layout

Restriction Rules

Scoping Rules

Object Access

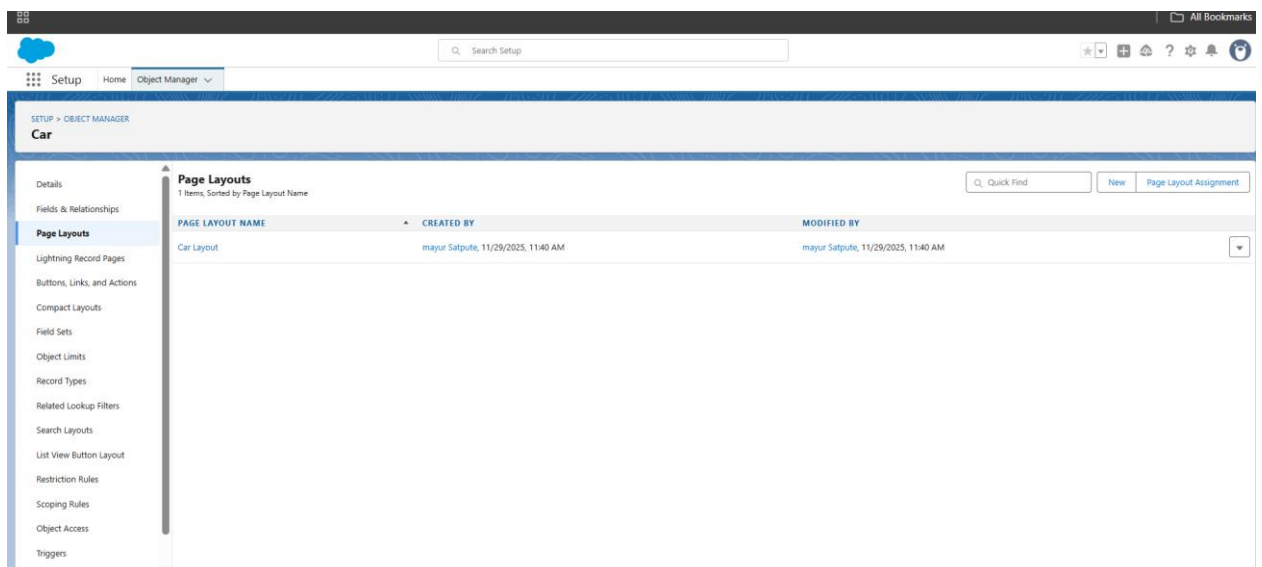
Record Typs

Daily Rental

- Long-Term Rental

Page Layouts

- `Car__c` page → Booking history
- `Car_Experience__c` page → Car + Customer info



Relationships

- Car → Booking: **Lookup**
- Booking → Contact: **Lookup**

Schema Builder

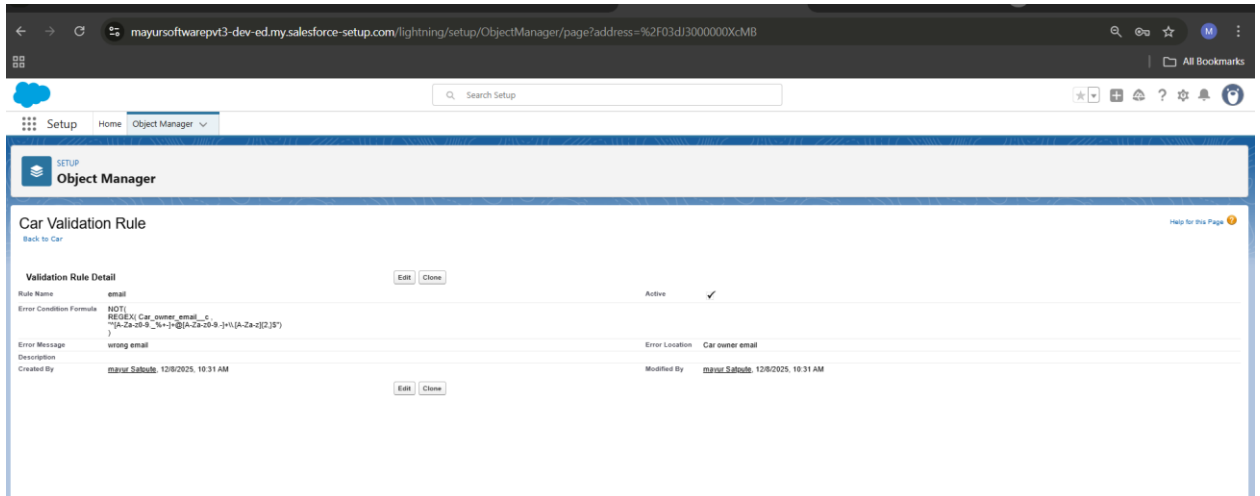
Visual mapping of object relationships.

Phase 4: Process Automation (Admin)

👉 **Goal: Automate tasks.**

Validation Rules

- End date must be after start date



Flows

- **Record-triggered Flow** → Calculate total amount
- **Screen Flow** → Booking form

Approval Process

- If `Total Amount > ₹50,000` → Manager approval

Email Alerts

- Send confirmation email

Field Updates

- After approval → Status = Confirmed

Tasks & Notifications

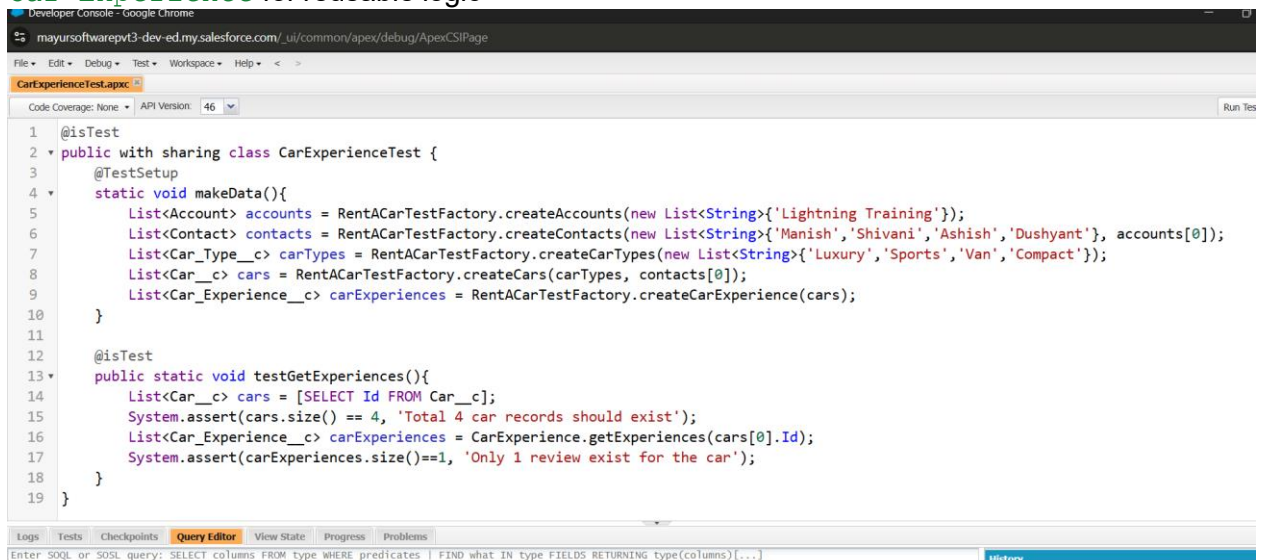
- Task for agent to prepare the car
- In-app notification after approval

Phase 5: Apex Programming (Developer)

👉 Goal: Write backend logic.

Classes

- `Car Experience` for reusable logic



```

1  @isTest
2  public with sharing class CarExperienceTest {
3      @TestSetup
4      static void makeData(){
5          List<Account> accounts = RentACarTestFactory.createAccounts(new List<String>{'Lightning Training'});
6          List<Contact> contacts = RentACarTestFactory.createContacts(new List<String>{'Manish','Shivani','Ashish','Dushyant'}, accounts[0]);
7          List<Car_Type__c> carTypes = RentACarTestFactory.createCarTypes(new List<String>{'Luxury','Sports','Van','Compact'});
8          List<Car__c> cars = RentACarTestFactory.createCars(carTypes, contacts[0]);
9          List<Car_Experience__c> carExperiences = RentACarTestFactory.createCarExperience(cars);
10     }
11
12     @isTest
13     public static void testGetExperiences(){
14         List<Car__c> cars = [SELECT Id FROM Car__c];
15         System.assert(cars.size() == 4, 'Total 4 car records should exist');
16         List<Car_Experience__c> carExperiences = CarExperience.getExperiences(cars[0].Id);
17         System.assert(carExperiences.size()==1, 'Only 1 review exist for the car');
18     }
19 }
  
```

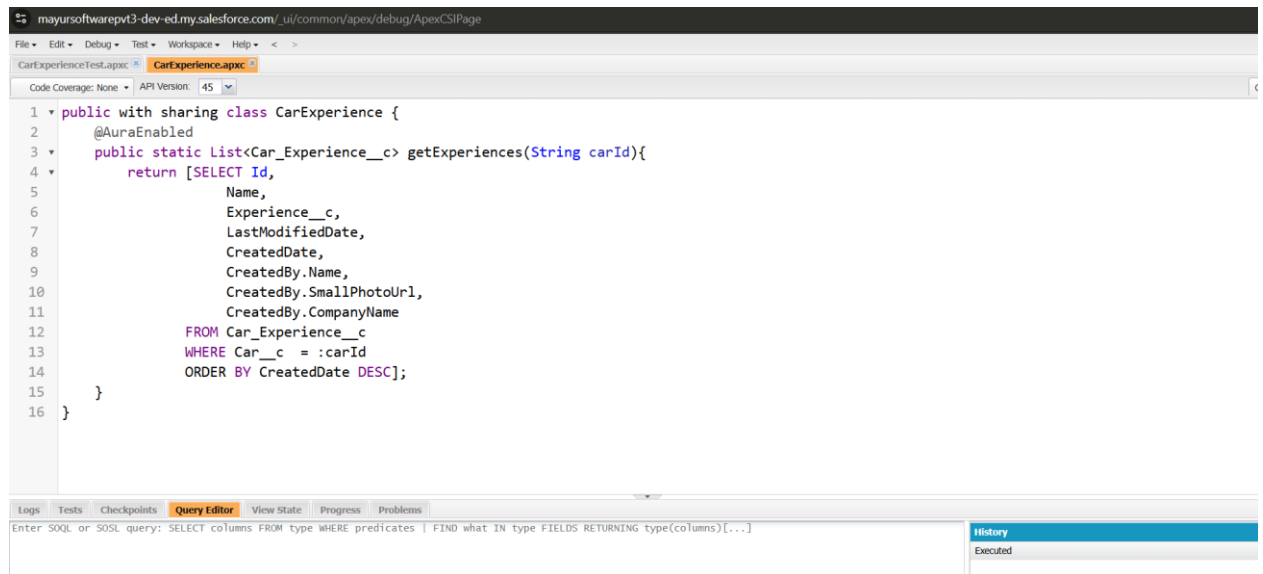
Enter SOQL or SOSL query: SELECT columns FROM type WHERE predicates | FIND what IN type FIELDS RETURNING type(columns)[...]

Triggers

- Prevent overlapping bookings on same car

SOQL

- Query available cars



```
1 public with sharing class CarExperience {
2     @AuraEnabled
3     public static List<Car_Experience__c> getExperiences(String carId){
4         return [SELECT Id,
5                 Name,
6                 Experience__c,
7                 LastModifiedDate,
8                 CreatedDate,
9                 CreatedBy.Name,
10                CreatedBy.SmallPhotoUrl,
11                CreatedBy.CompanyName
12                FROM Car_Experience__c
13                WHERE Car__c = :carId
14                ORDER BY CreatedDate DESC];
15     }
16 }
```

Enter SOQL or SOSL query: SELECT columns FROM type WHERE predicates | FIND what IN type FIELDS RETURNING type(columns)[...]

History
Executed

Collections

- Use List, Set, Map to handle multiple records

Batch Apex

- Mark overdue rentals night job

Queueable Apex

- Apply bulk discounts asynchronously

Scheduled Apex

- Daily rental summary email to manager

Future Methods

- Insurance API callouts

Test Classes

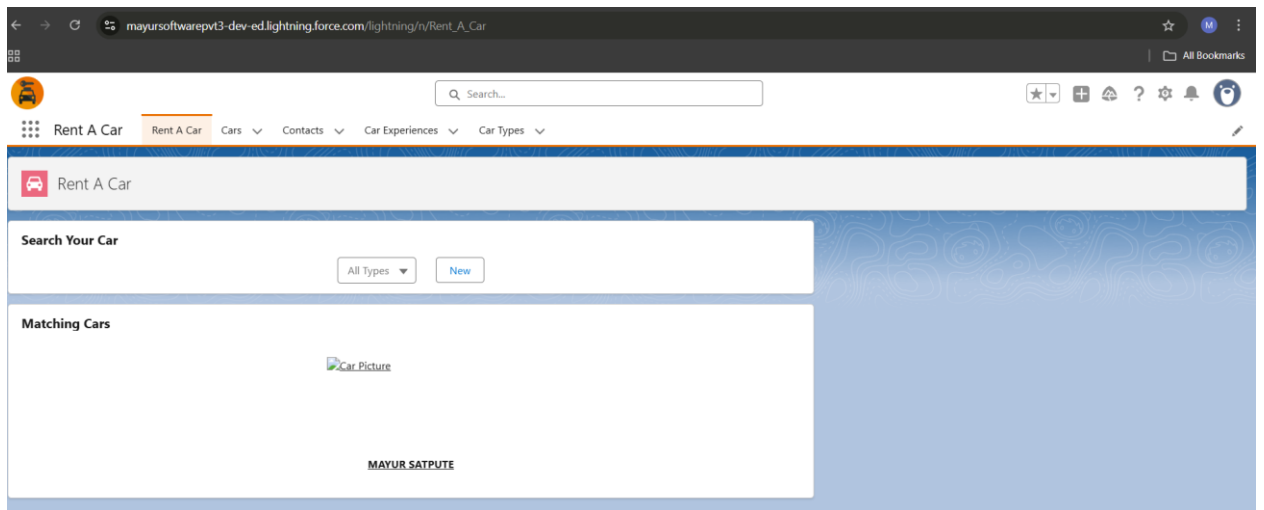
- Create test data
- Validate trigger logic

Phase 6: User Interface Development

 **Goal: Build user-friendly UI.**

Lightning App Builder

- Create “Car Rental CRM” app



Record Pages

- Car page with related bookings

Tabs

- Car
- Booking

Home Page

- Fleet utilization dashboard

Utility Bar

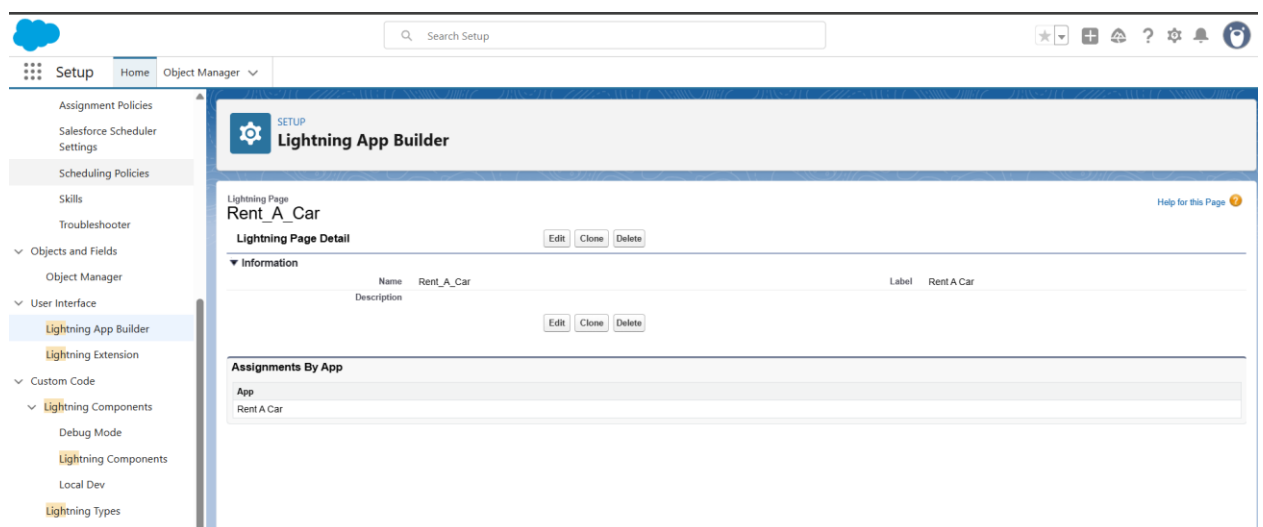
- Quick “New Booking” button

Lightning Web Components

- Search available cars by date
- Display in datatable
- Book Now → Apex call

Navigation Service

- Redirect to booking record



Phase 7: Integration & External Access

👉 **Goal:** Connect external systems.

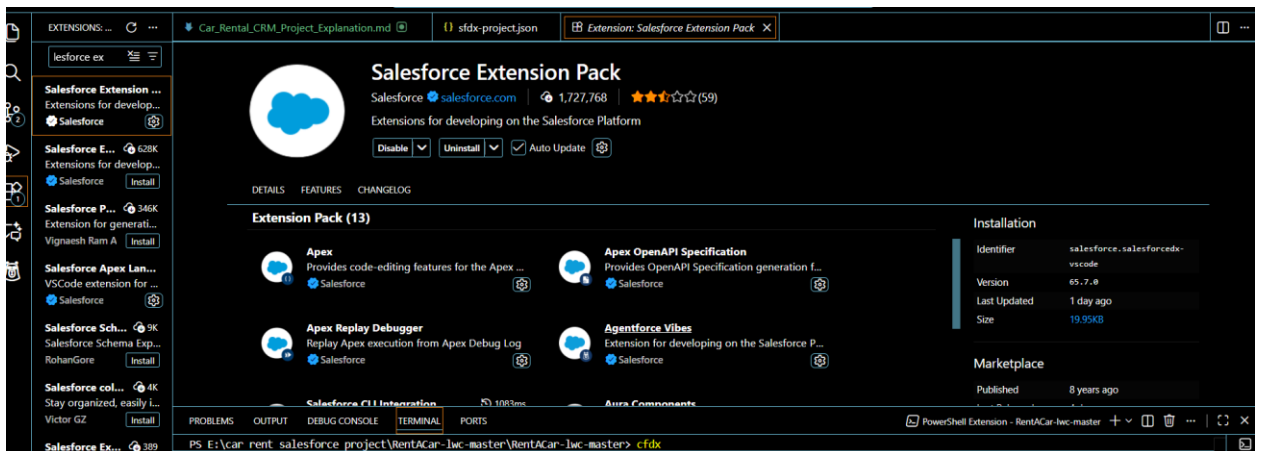
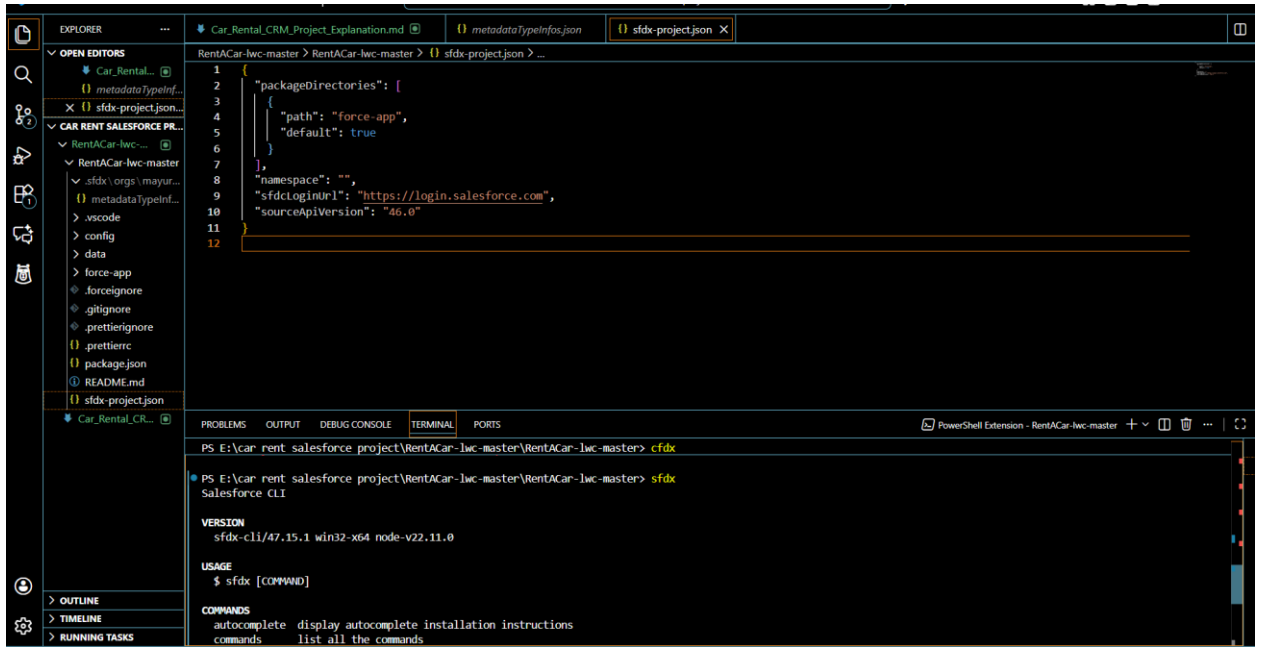
- **Named Credentials** → Store API keys securely
- **External Services** → Car insurance verification

- **REST/SOAP Callouts**
 - **Platform Events** → Car breakdown alerts
 - **Change Data Capture** → Notify external apps
 - **Salesforce Connect** → Link external DB
 - **Remote Site Settings** → Allow API domains
 - **OAuth** → Customer login
-

Phase 8: Data Management & Deployment

👉 **Goal: Manage and deploy data.**

- **Data Import Wizard**
- **Data Loader**
- **Duplicate Rules**
- **Weekly Backups**
- **Change Sets**
- **Managed vs Unmanaged Packages**
- **ANT Migration Tool**
- **VS Code + SFDX**

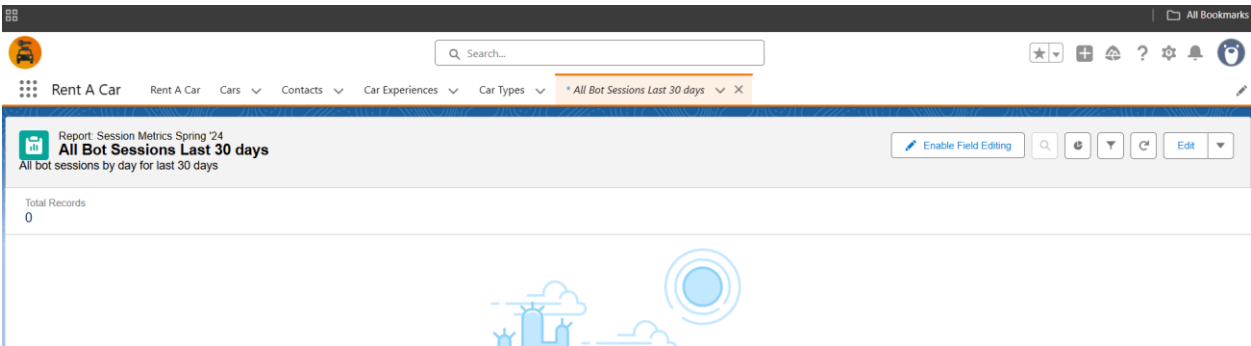


Phase 9: Reporting, Dashboards & Security Review

👉 Goal: Analyze & protect business data.

Reports

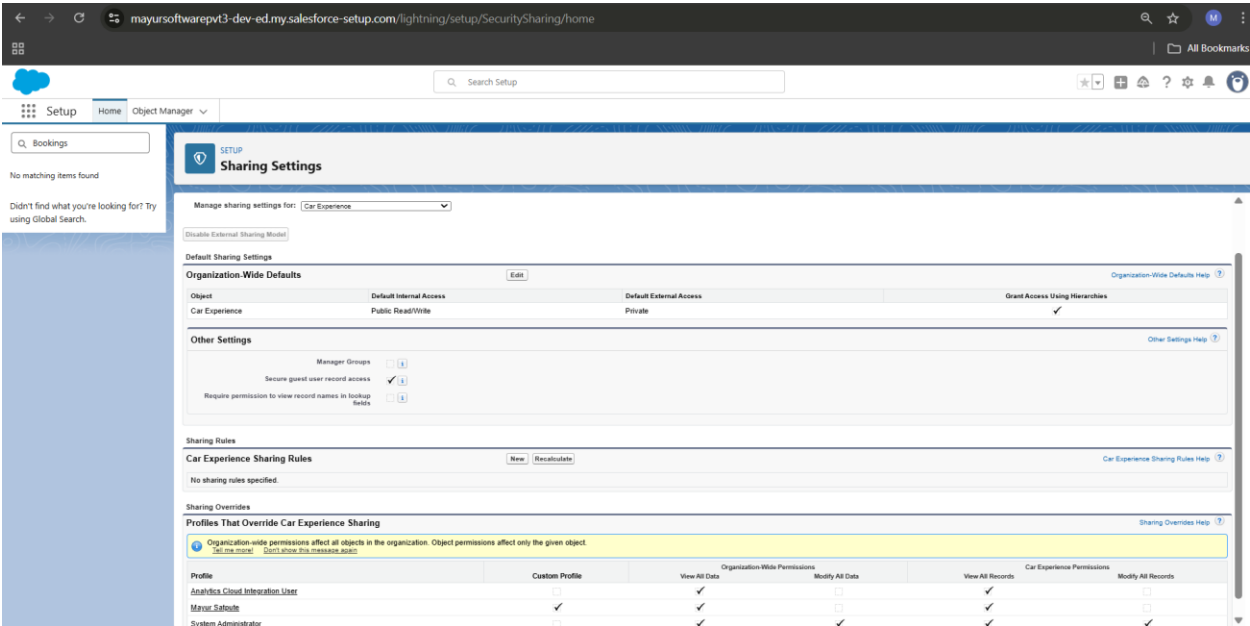
- Car Utilization Report
- Revenue by Car Model



Dashboards

- Fleet Utilization Dashboard
- Manager Revenue Dashboard
-

Security Settings



- Cars → Public
- Field Level Security
- Login IP Ranges
- Audit Trail

Date	User	Source Namespace Prefix	Action	Section	Delegate User
12/8/2025, 10:31:45 AM PST	mayur.satpute@gmail.com		New Cars validation rule "email"	Validation Rules	
12/8/2025, 10:29:47 AM PST	mayur.satpute@gmail.com		Changed field name of Car owner email custom field on Cars from Car_owner_email to Car_owner_email	Custom Objects	
12/8/2025, 10:29:47 AM PST	mayur.satpute@gmail.com		Changed label of custom field on Cars from Car owner email to Car owner_email	Custom Objects	
12/8/2025, 10:28:10 AM PST	mayur.satpute@gmail.com		Changed Car page layout Car Layout	Custom Objects	
12/8/2025, 10:28:10 AM PST	mayur.satpute@gmail.com		Created custom field Car owner email (Text) on Cars	Custom Objects	
12/8/2025, 10:23:38 AM PST	Automated Process		Removed the UserInteractionInsights user permission as allowed by User License 1005000001eY7b		
12/4/2025, 9:03:47 PM PST	mayur.satpute@gmail.com		Enrolled mayur.satpute@gmail.com in Lightning Login	Manage Users	
12/4/2025, 9:03:23 PM PST	salesforce.com, inc.		Max number of streaming topics		
11/29/2025, 11:49:14 AM PST	mayur.satpute@gmail.com		Permission set RentACar assigned to user mayur Satpute (UserID: 00550000039kngj)	Manage Users	
11/29/2025, 11:49:40 AM PST	mayur.satpute@gmail.com		Changed tabs of custom app Rent A Car	Custom Apps	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Mailing Address was changed from No Access to Read/Write	Manage Users	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Email was changed from No Access to Read/Write	Manage Users	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Data.com Key was changed from No Access to Read/Write	Manage Users	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Mobile was changed from No Access to Read/Write	Manage Users	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Reports To was changed from No Access to Read/Write	Manage Users	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Fax was changed from No Access to Read/Write	Manage Users	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Do Not Call was changed from No Access to Read/Write	Manage Users	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Other Address was changed from No Access to Read/Write	Manage Users	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Assistant was changed from No Access to Read/Write	Manage Users	
11/29/2025, 11:40:47 AM PST	mayur.satpute@gmail.com		Changed permission set RentACar: field-level security for Contact: Home Phone was changed from No Access to Read/Write	Manage Users	

Phase 10: Final Presentation & Demo Day

👉 Goal: Present your project professionally.

Github Link

https://github.com/satputemayur/Car_Rent-CRM-Salesforce-Project

Demo

<https://drive.google.com/file/d/15qKj2rAYFcggMR8ut2ekPUPIsDfEDb8x/view?usp=sharing>